Notes about multimedia forensics

Digital Forensics and Biometrics 2024/2025 – Simone Milani

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Introduction

The widespread availability of inexpensive, user-friendly devices enables almost everyone to record, store, and share digital documents, messages, images, videos and audio files. Sat the same time, accessible and easy-to-use editing tools make the alteration and editing of such contents a relatively-simple and immediate task without the need for a visual editing expert or audio engineer. In addition, current computer graphics or AI-enabled software can produce photorealistic graphics and hybrid images that could result perceptually indistinguishable from photographs acquired from a real scene.

As a result, digital images/audio/videos often undergo various processing stages, ranging from simple quality enhancement and compression to content alterations and composition with other preexisting materials. The recent appearance of editing tools based on deep learning has led to a rise in doctored or totally-synthetic images, eroding trust in visual content and the assumption that "seeing is believing." Indeed, Europol forecast that by 2025 90% of the contents on the internet will be generated or edited with the use of AI tools.[1]

This growing challenge highlights the need for methods to trace an image's history to verify its authenticity and quality. As a matter of fact, several questions arises such as:

1) **who captured or created the content?** Often by examining metadata, it is possible to obtain information about the owner. Despite this, such traces can be easily erased and therefore, forensic experts need to resort to or device-specific traces.
2) **What is the origin of such content?** The authenticity or reliability of the media can be verified the claimed source, such as the device or software used.
3) **Was it modified? i.e., wat type and extent of changes were made?** In this analisys, experts want to verify additions, deletions, or enhancements that were made to the contents, verifying **how much of the original content remains intact**.
4) **Can the content be trusted?** Editings may change the semantic meaning and the information that is displayed/reported by the multimedia content (e.g., erasure in an eavesdropped audio sequence).

In addition, multimedia contents employed as evidences of proves imply a set of additional questions such as

1) **What this multimedia content tells me about the scene where it was acquired?** In this context, the analysis focus on extracting information about the real scene where it was captured, such as environmental details, objects, or events. The main aim is to prove or verify some assumptions or hypothesis about the recorded event.
2) **Is it possible to relate it to other contents?** Many investigations involves multiple heterogeneous evidences that need to be properly linked and ordered, e.g., cross-

[1] Europol Innovation Lab (2021). Facing Reality: Law Enforcement and the Challenge of Deepfakes. https://www.europol.europa.eu/cms/sites/default/files/documents/Europol_Innovation_Lab_Facing_Reality_Law_Enforcement_And_The_Challenge_Of_Deepfakes.pdf

referencing timestamps, geolocation tags, or related media files. This helps establish a timeline or connections and provide a more detailed overview of the events.

3) **Was the software/device who generated/took it legal?** The set of activities answering such questions aim at assessing the legality of the software or device used to capture or generate the contents. These include verifying whether the device is licensed, the software is authorized, and if it complies with established regulations, which is crucial in legal or forensic applications.

To address these issues, researchers in multimedia content security have developed **active** and **passive** approaches. *Active methods* rely on data recorded during the image's acquisition (e.g., camera metadata), while *passive method* rely on the observation that each phase of the image history, from the acquisition process, to its storing in a compressed format, to any post processing operation leaves a distinctive trace on the data, as a sort of digital fingerprint.
It is then possible to identify the source of the digital multimedia content or determine whether it is authentic or modified by detecting the presence, the absence, or the incongruence of such features.

Image forensics

The lifecycle of a digital image

The history of a digital image is a sequence of processes that can be categorized into three main phases: **acquisition**, **image editing and processing**, and **compression**. The second phase can be divided into **in-camera processing** and **post-acquisition processing**. Each phase contributes distinct characteristics and leaves behind identifiable traces, which can be analyzed for forensic purposes.

1. Acquisition phase: capturing the image
At this stage, a real-world scene is recorded into a digital picture through the camera's imaging system, whose structure can be modelled by a pipeline of subsequent processing stages::

- **Image focusing:** incoming light, which has been generated by the radiation emitted by a light source and reflected by the objects of the scene, is captured by an optical lens that make rays collimated on a photo responsive plane. Lenses consist of multiple glass or plastic elements (depending on the cost and technical level of the device) that are arranged in a specific configuration to control the direction and quality of incoming light. The **focal length** of a lens determines the ability of the lens in acquiring in-focus objects and define its field of view: shorter focal lengths (wide-angle lenses) capture broader scenes, but longer focal lengths (telephoto lenses) magnify distant subjects. Most lenses enable adjustable **apertures** to regulate the amount of light entering the camera and control the depth of field (the range of possible distances where objects can be acquired in focus). Other features may include **optical zoom**, which changes focal length without compromising image resolution, and **image stabilization**, where blurring due to camera shake is minimized. Lenses also contribute significantly to image quality, affecting sharpness, distortion, and chromatic aberration, making them integral to capturing clear and accurate digital photographs.
- **Light capture**: Light from the scene is focused onto the camera sensor (usually a CCD or CMOS) by the camera's lens system. This initiates the creation of the digital image signal. Such sensors are made of matrices of photoresponsive elements corresponding to the image pixels: each sensor captures incoming light and convert its intensity (a.k.a. number of photons) into a charge or tension value. Reading can instantaneous or sequential; indeed, the behavior of these elements if very similar to buckets collecting rain (photons) until

a certain level (the pixel value) is reached. Note that the acquired value depends on the amount of acquired photons (energy) independently from their wavelength (color). As a matter of fact, it is necessary to acquire color information using a set of frequency-selective filters within the visible light spectrum.

- **Color Filtering**: Before reaching the sensor, the light passes through a *Color Filter Array (CFA) or Bayer mask*, a thin film that selectively allows only specific wavelengths of light corresponding to one primary color—Red, Green, or Blue—to reach each sensor pixel. This filtering means that at this stage, each pixel captures only one primary color component. This process ensures that each pixel can be associated to three color components after an interpolation operation (see later points).

2. In-camera processing

- **Demosaicing**: To reconstruct a full-color digital image, the single-color data for each pixel is interpolated. This process, known as **demosaicing**, estimates the missing color components (Red, Green, or Blue) for each pixel to produce a complete color image.

Once the raw image data is reconstructed (3 colors per pixel), it undergoes additional processing within the camera. These steps include:

- **White balancing**: adjusts color tones to achieve accurate color reproduction; indeed white balancing ensures that whites appear neutral, compensating for the color temperature of the light source. Different lighting conditions, such as sunlight, tungsten, or fluorescent light, emit varying hues that can affect the overall color of a photo. The camera analyzes the scene to determine the dominant light source and shifts the colors accordingly.
- **Color processing**: enhances or standardizes color information; in this phase, color saturation can be enhanced, and color casts are corrected. Additional adjustments, such as tone mapping could be applied for a better final picture.
- **Image sharpening**: emphasizes edges and enhances the clarity and definition of details. To this purpose, sharpening algorithms identify transitions between different tones or colors, such as the boundary between an object and its background. By augmenting the contrast along these edges, images appears crisper and more defined making fine details more evident.
- **Contrast enhancement**: adjusts light and dark regions for better visual appeal. In this case, camera firmware adopts a tone mapping strategy to adjusts the difference between the lightest and darkest areas to improve visual clarity and depth. This can involve stretching the dynamic range to make shadows darker and highlights brighter or selectively enhancing midtones for a balanced look. Some cameras use local contrast enhancement to boost detail in specific regions without overexposing or underexposing others.
- **Gamma correction**: modifies the brightness to align with human perception. The raw data captured by the camera sensor are linear, i.e., brightness values are proportional to light intensity. Human vision is instead more sensitive to changes in dark tones than in bright ones. Gamma correction applies a mathematical curve to redistribute these brightness levels, making dark areas more visible while preventing highlights from becoming overly bright.

Note that some cameras enable acquiring images in RAW file format, which reports pixel values after photo-sensors acquisition (i.e., without applying white balancing, color and gamma corrections, sharpening or enhancement). In this way, all the previously-described operation can be performed with photo-editing software and customized for that specific picture.

3. Coding phase: storing images

Once the image signal is processed, it is stored in the camera's memory. Compression strategy are usually divided into **lossless** and **lossy** strategies. *Lossless strategies* aim at compressing the image (or video or audio) file without losing any information: when decoding, the codec software is able to reconstruct the original image perfectly with no alteration or artifacts. *Lossy strategies* instead aim at achieving high compression rations by introducing some distortion on the reconstructed image. In this case, operations like quantization and zeroing allow simplifying the coded signal thus reducing the coded bit rate (i.e., enhancing the compression ratio) at the cost of some additional coding distortion affecting the reconstructed content. Such operations enable a bit rate reduction since redundant or less noticeable details are removed. Since most of these steps are **non-normative** (i.e., standards does not provide strict guidelines on how to implement them, and therefore, different device vendors or platforms adopt their own strategies – usually patented - in order to make the image looks better), it is possible to take over the related compression artifacts to trace back the origin of the content.

Examples of lossless compression formats: PNG, BMP, JP2000 (jpeg2000 codec), GIF
Examples of lossy compression formats: JPEG, GIF and JP2000 (with some configurations), WebP, DjVu, PGF.

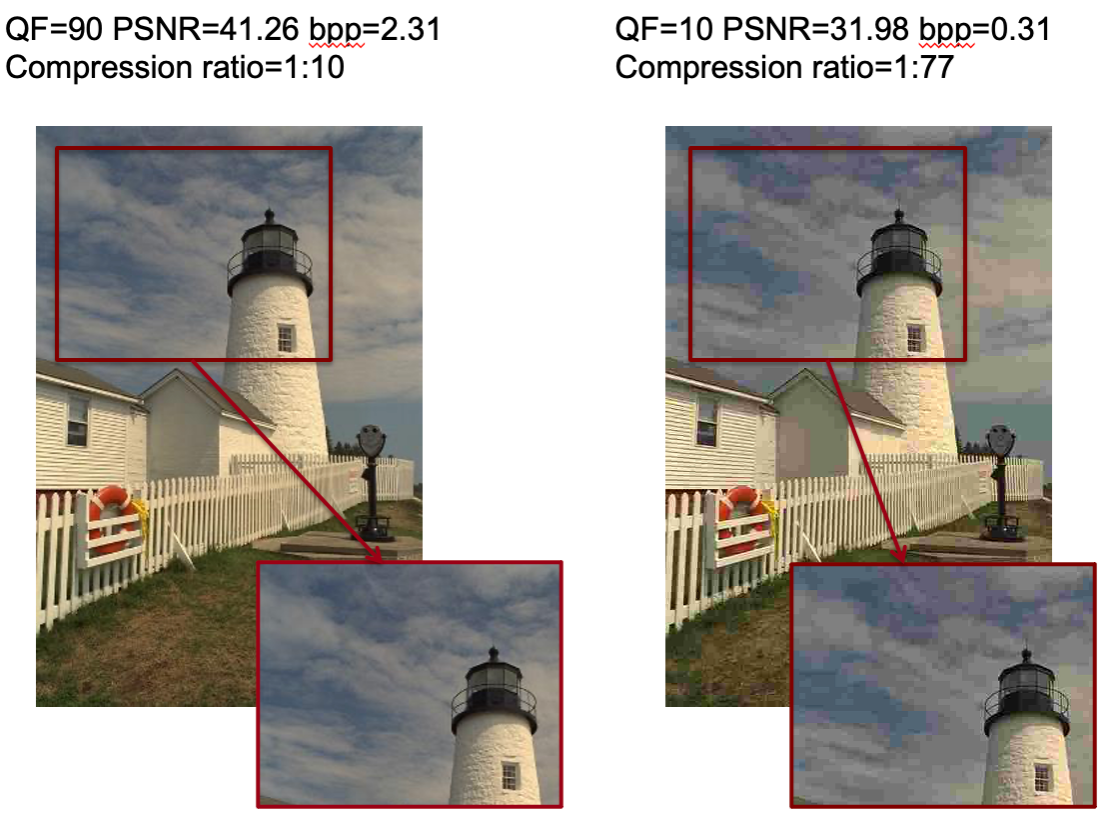


Figure 1Example of different compression levels for JPEG codec.

In the end, we can say that most of the images available online are in **JPEG** format.

When taling about video contents, most of the files are compressed using lossy mode (because of the large amount of storage that would be required to store a RAW video sequence). Among the possible standards, we can mention MPEG1.
,2,4, H.264/AVC, H.265/HEVC, H.266/VVC, H.264, H.264/SVC.

As for audio tracks, WAV files are lossless coded (CD quality), while formats like MP£ or OGG are lossy.

The main operations in lossy image codecs imply representing the input signal with respect to an efficient representation basis (.e., input data are transformed into a set of coefficients such that the original signal can be reconstructed as a linear combination of basis elements weighted by the coefficients). Such transformations are usually invertible and therefore, transforming data does not imply information loss. One of the most widely used transforms for image compression is the Discrete Cosine Transform (DCT), that can be written for image blocks of NxN pixels as

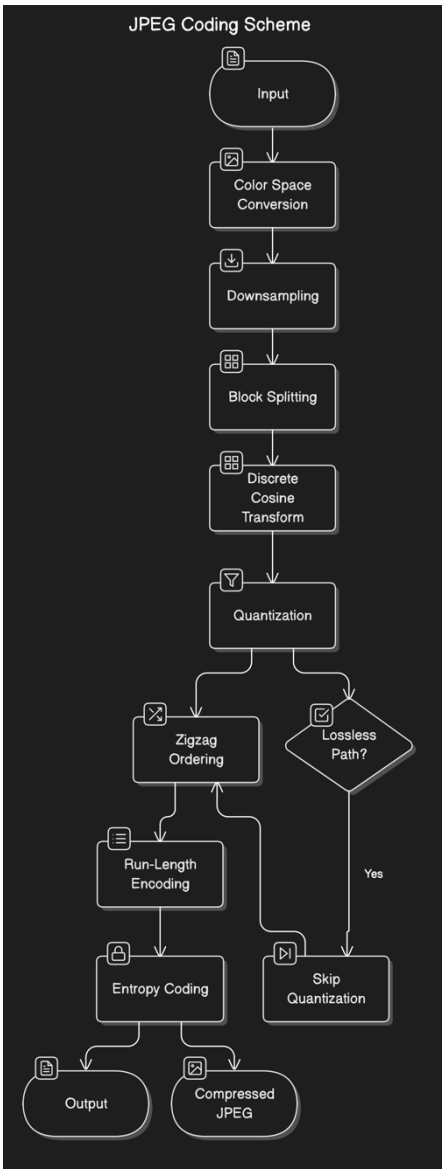
$$F[i,j] = \frac{1}{4} C(i) C(j) \sum_{x=0}^{7} \sum_{y=0}^{7} P[x,y] \cos \frac{(2x+1) i \pi}{16} \cos \frac{(2y+1) j \pi}{16}$$

$$C(h) = \begin{cases} 1, & h = 1,\ldots,7 \\ 1/\sqrt{2}, & h = 0 \end{cases}$$

where P[x,y] are pixel values and F[I,j] are frequency coefficients. Figure 2 reports an example of transformation operated on a matrix of 8x8 integers.

After transformation, coefficients are quantized with varying quantization steps (the frequency-varying pattern of quantization steps could be changing according to the codec or platform where image is uploaded). Quantized values are then coded into the bitstream.

As for JPEG image compression standard, it is possible to characterize the compression pipeline as reported in the block diagram on the right



Whenever referring to video compression, these two steps could be anticipated by a temporal-prediction that reduces temporal redundancy by specifying the innovation part of a frame (i.e., details that can not be found in previous frames). The predictable elements (e.g., motionless objects or elements that are moving according to a simple translational shift) could be estimated from previously-decoded frames by means of motion vectors.

75	76	77	78	80	81	82	83
76	76	77	79	80	82	83	83
77	77	78	80	81	83	84	84
78	79	80	81	83	84	85	86
80	81	82	83	85	86	87	88
82	82	83	85	86	88	89	89
83	83	84	86	87	89	90	90
83	84	85	86	88	89	90	91

DCT

664	-22	0	0.39	0	0.12	0	-0.17
-23.8	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1.03	0	0	0	0	0	0	0
0	0.15	0	-1.03	0	-1.54	0	0.75
-0.31	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0.19	0	0	0	0	0	0	0

Figure 2 Example of DCT trasform applied on an 8x8 block of pixels

4. Editing Phase: Post-Processing the Image

After the image is created, it can undergo various post-processing modifications. These edits might be intended for improvement or manipulation of the content and can be performed directly on the acquiring device (i.e., retouching performed directly with smartphone software or camera firmware) or on different computers/servers (e.g., during uploads or photo-editing the contents before distribution). Common editing operations include:

- **Geometric transformations**: Adjusting the image through rotation, scaling, or cropping.
- **Blurring and sharpening**: Smoothing or enhancing image details.
- **Contrast adjustment**: Modifying brightness or contrast to alter the image's appearance.
- **Image splicing**: Composing a new image by combining parts of multiple images.
- **Copy-move (cloning)**: Replicating sections within the same image to hide or emphasize certain areas.
- **Recompression**: whenever a compressed image (e.g., JPEG) is edited, processing software need to reconstruct it since it operates in the pixel domains. After such editings, altered images are saved back again in the JPEG format, introducing further compression artifacts.

Being many of the previous editings not reversible, it is possible to detect editing traces left on the image signal in order to identify the processing history of the content or the device that acquired/processed it (by identifying the specific algorithm).

Forensic digital image fingerprints

As previously stated, the processes leave behind unique, detectable traces, referred to as **digital fingerprints** or **footprints**, which can be analyzed for forensic purposes. These include:

1. **Acquisition Fingerprints**: Traces from the sensor (e.g., CFA pattern) and in-camera processing (e.g., demosaicing algorithms or sharpening techniques).
2. **Coding Fingerprints**: Artifacts introduced during image compression and storage (e.g., JPEG quantization patterns).
3. **Editing Fingerprints**: Evidence of manipulations such as splicing, cloning, or recompression.

By analyzing these fingerprints, forensic investigators can reconstruct the history of a digital image to determine its authenticity, detect modifications, and identify the processes it underwent during its lifecycle. This method is fundamental in understanding and verifying the credibility of digital visual content.

Lens-related traces

Each camera model has unique lens characteristics due to variations in design and manufacturing processes. Lenses introduce specific types of aberrations (such as chromatic aberration, vignetting, and lens distortion) leaving distinctive traces that can be analyzed to link an image to a particular device (or detect tampering).

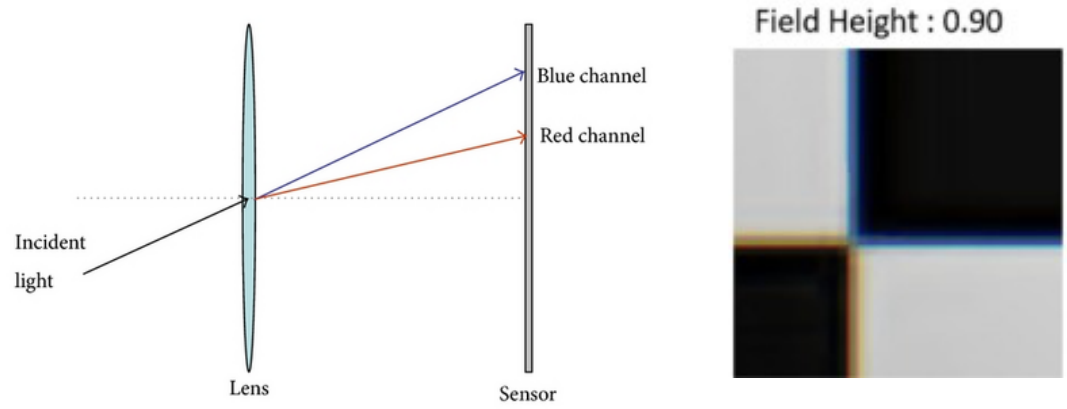


Figure 3 Example of lateral chromatic aberration

One such aberration is **lateral chromatic aberration**, which occurs when light wavelengths focus on slightly different points in the image plane, especially for light entering the lens off its optical axis. The result is a color fringing around high-contrast edges due to the lens's inability to make all wavelength converge (see the following image). By assuming that this aberration remains consistent within each color channel, the shifts between the red-green and blue-green channels are estimated, often using the green channel as a reference. The aberration is modeled using three parameters: two for the distortion center and one for its magnitude. Detecting tampering involves identifying local inconsistencies in these parameters compared to the overall image.
This concept has been adapted for various forensic applications. For instance, algorithms have been refined to use these distortion parameters with machine learning models, like support vector machines (SVMs), for identifying the source device.

Other methods exploit **radial distortion** caused by lens curvature or even dust patterns, modeled with Gaussian functions, to pinpoint a specific device. Radial distortion arises from the spherical shape of the lenses, which causes light to be refracted differently between its

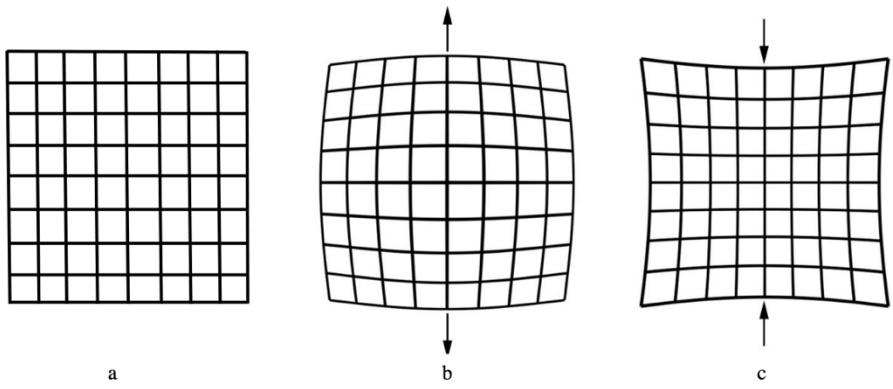


Figure 4 Example of radial distortion

center and edges. Straight lines are transformed (in an exaggerated way) into arcs of circles, drawing either a “barrel” or a “pincushion” shape (see figure 4). This effect is more pronounced near the periphery of the lens and is influenced by the optical design, particularly the curvature of the lens elements. Accurate calibration and correction algorithms, such as those employing polynomial models or division models, are commonly used in photogrammetry and computer vision to mitigate the impact of radial distortion on imaging accuracy. Such estimation parameters could be used as distinctive elements for lens identification. As an example, representing a pixel in radial coordinates, i.e., $(x, y) = r \cdot e^{-j\,\theta} + (x_0, y_0)$ with respect to the image center (x_0, y_0), it is possible to compensate the distortion on r with the equation

$$r_d = r_u + k_1 \cdot r_u^3 + k_2 \cdot r_u^5$$

Where suffix d denotes “distorted” and suffix u denotes “undistorted”. When parameter k_1 is lower than 0, we have pincushion; barrel distortion for $k_1 > 0$.[2]

Additionally, **purple fringing aberration (PFA)**—a stronger, more visible effect causing blue-purple halos around high-contrast edges—is used for tampering detection.

The same footprints can be applied to other optical-related devices, such as scanners, where additional traces can be taken into account. Lenses can be found in other devices like printers, although different types of traces can be taken into account for their identification (see appendix at the end of the notes).

Sensor-related traces

A short introduction on noise sources in digital camera sensors

As previously-mentioned, circuitry and sensing image samples are affected by different types of noises, among which we can mention:

- **Photon Shot Noise**, i.e, a stochastic variation of the measured pixel value due to the quantum nature of light and random arrival of photons at the sensor. Such noise signal can be modelled by a Poisson distribution (modelling arrival of photons), i.e.,

  $p(n) = \frac{\lambda^n}{n!} e^{-\lambda}$ for $n \in \mathbb{N}$

  And can be approximated by Gaussian noise for large numbers. Such noise dominates in low-light conditions where fewer photons are available, and is proportional to the square root of the light intensity. As a matter of fact, the Signal-to-Noise Ratio (SNR) can be written as

[2] K. S. Choi, E. Y. Lam, and K. Wong. Source camera identification using footprints from lens aberration. In Proc. SPIE, Electronic Imaging, Media Watermarking, Security, and Forensics, volume 6069, pages 60690J–60690J–8, 2006.

$$SNR = \sqrt{N}$$

where N is the average number of events.

- **Dark current noise**: these fluctuations of pixel values are due to the thermal generation of electrons within the sensor's photodetector array, even in the absence of light. Such noise is temperature-dependent (as it increases with higher operating temperatures) and spatially-variable (since its is affected by non-uniform sensor fabrication characteristics). As a result, image display glows or uneven dark patterns in long exposures due to the reported oscillations. Despite being intrinsic to the fabrication characteristics, such noise is not systematic and varies from picture to pictures.
- **Fixed Pattern Noise:** the origin of FPN are variations in the physical properties of pixels and imperfections in sensor manufacturing. Such signal is visible as a static texture overlaying the image, and can be divided into two components: **dark signal non-uniformity** of **DSNU** (which is more complex to estimate) and the **photoresponse non-uniformity** or **PRNU** (which will be described in the detail, within the next section).
- **Temporal noise**: this is characterized by random fluctuations in pixel signals between consecutive image captures.
- **Readout noise**: it is introduced during signal conversion and amplification by the sensor's readout electronics.
- **Thermal noise**: similar to DCN, it is generated by the thermal agitation of electrons, which can be more evident whenever adopting a long exposure time.
- **Cross-talk noise**: such alterations are due to the proximity interference effects (optical or electrical) between adjacent pixels.
- **Amplifier noise**: whenever pixel values are amplified and ranges are stabilized, electronic equipment could introduce some extra disturbances.

From this overview, it is possible to notice that PRNU correspond to a fixed pattern which proves to be distinctive of the adopted device.

Photo Response Non-Uniformity (PRNU) noise

Photoresponse Nonuniformity (PRNU), which is a high-frequency multiplicative noise present across the sensor array. PRNU remains stable under normal operating conditions and is uniquely identifiable for each camera, making it a vital tool for forensic applications such as device identification, single-device linking, and forgery detection.

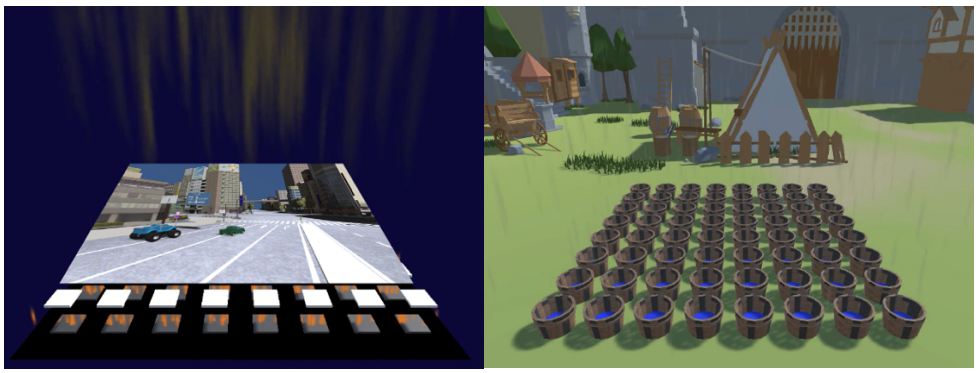


Figure 5 Photon acquisition process for an image

It is possible to consider the process of light intensity acquisition operated by the photo-responsive sensor s of a digital camera similar to collecting rain using a regular array of buckets. In case the rain (photons) falling is uniform and the size of each bucket is identical,

the amount of collected water (pixel intensity) will be the same. In case buckets show different in size and aperture, water amounts will be varying according to bucket's characteristics.

In a similar way, photosensors presents several differences due to:

- *Manufacturing imperfections,* i.e., variations in doping levels and sensor pixel construction.
- *Silicon inhomogeneities*, due to microscopic differences in the silicon substrate affecting light absorption.
- *Thermal noise,* which leads to minor variations (with respect to the other factors) and depends on the operational heat.

Mathematical modelling of PRNU noise

The image sample (pixel) acquired by a digital camera can be modelled as the combination of several factors. From a mathematical point of view, it is possible to write that pixel intensity I is

$$I = I_0 + K\ I_0 + \Theta. \quad (*)$$

where I_0 is the noise-free sensor output (i.e., the pixel value with no noise), K is the PRNU noise for the current camera (i.e., a zero-mean multiplicative noise related to the manufacturing process and depending on the intensity of the incoming light), and Θ includes all the others noise terms (shot, readout, etc.) that are usually modelled by a Gaussian distribution (i.i.d.).

A more detailed modelling includes some gain and non-linear rescaling elements that lead to the extended equation

$$I = g^{\gamma}\ [I_0 + K\ I_0 + \Theta]^{\gamma} + Q$$

where g is a gain factor and Q refers to compression noise.

It is possible to isolate the PRNU component by denoising the acquired image I using a denoising filter (e.g., Wiener) or any denoising function/networks (consider the case of DnCNN, largely used in many multimedia forensics applications). We can call the denoised image $\hat{I} = F(I)$, where F is the chosen denoising operator.

Given $\hat{I}$, it is possible to compute the residual $W_I = I - \hat{I}$ that can be approximated by the sum of a multiplicative component and a Gaussian noise, i.e., $W_I = I - \hat{I} = IK + \Sigma$.

As a verification for this, consider

$W_I = I - \hat{I} =$ (add and subtract IK) $= I - \hat{I} + IK - IK =$ (replace I with *) $= I_0 + K\ I_0 + \Theta - \hat{I} + IK - IK = IK + I_0 - \hat{I} + K(I_0 - I) + \Theta =$ (note that if denoising operator F is effective we can assume $F(I) = \hat{I} \simeq I_0$) $= IK + 0 + K(I_0 - I) + \Theta =$ (being $I_0 - I$ very small, like K, we can gather $K(I_0 - I) + \Theta = \Sigma$, which is independent from IK) $= KI + \Sigma.$

This operation can be repeated for several images $I_1, I_2, \dots, I_N$, and the final $\widehat{K}$ can be computed as

$$\widehat{K} = \frac{\sum_{i=1}^{N} W_i I_i}{\sum_{i=1}^{N} (I_i)^2}$$ (maximum likelihood estimator)

This process is repeated for multiple devices to create a database of PRNU patterns for identifying and linking cameras.

After such estimation, it is possible to verify if an image J can be well-matched to an estimated PRNU noise $\widehat{K}$ by denoising the image J in $\hat{J} = F(J)$ and computing the difference $W_J = J - \hat{J}$. Then, correlation between W_J and $\widehat{K}J$ can be computed using two possible metrics

- NCC (Normalized Cross-Correlation)

$$\rho = NCC(X, Y) = \frac{(X - \bar{X}) \cdot (Y - \bar{Y})}{\|X - \bar{X}\| \cdot \|Y - \bar{Y}\|}$$

  where $\|X\| = \sqrt{X \cdot X}$ and $X \cdot Y = \sum_{i,j} X[i,j]Y[i,j]$.

- PCE (Peak-to-Correlation Energy), that can be computed as

$$PCE = \frac{\rho^2}{\frac{1}{MN}\sum_i \sum_j NCC_{i,j}}$$

  where

$$NCC_{i,j} = \frac{\sum_{k=0}^{M-1}\sum_{l=0}^{N-1}(X[i,j]-\bar{X})\cdot(Y[k+i,l+j]-\bar{Y})}{\|X-\bar{X}\|\cdot\|Y-\bar{Y}\|}$$

Some hints for a good PRNU estimation:
▪ use flat-field images (uniform content, low variance)
▪ the luminance of the images should be as high as possible but not saturated
▪ about 20 flat-field (bright) images suffice to have a good estimate of K.

Applications of PRNU
This type of noise can be used for multiple forensic applications.
1. **Device Identification**: PRNU serves as a fingerprint to link an image to its source camera, enabling tracking across images.
2. **Forgery Detection**: Regions within an image with inconsistent PRNU patterns can reveal tampering, such as replacing parts of the image with content from a different device. A block-wise two-hypothesis test (tampered/non-tampered) is performed to assess integrity.

Challenges and Enhancements

Despite its efficiency and large use, PRNU estimation presents several problems.

Robustness problems:
To improve the reliability of PRNU extraction, you can use advanced methods enhance noise residuals while preserving the PRNU signal (better denoising).
In addition, PRNU estimation can focus on high Signal-to-Noise Ratio (SNR) regions: this reduces interference from high-frequency image content.
Finally, since strong compression erases most of the PRNU traces, it is important to operate on mildly-compressed or uncompressed data.

Firmware-related traces

Color interpolation traces

Digital cameras acquire images by converting light into grayscale data using sensors such as CCD or CMOS arrays. To reproduce colors, additional processing is required since each sensor can acquire only light intensity. For these reasons, digital camera sensors can be divided into two main categories.

3-chip color sensors. These systems use prisms and filters to separate incoming light into red, green, and blue (RGB) components, directing them to three distinct sensors for acquisition.

Single-chip color sensors. They employ color filter arrays (e.g., Bayer masks) to capture RGB data. Reconstruction of color images occurs through demosaicing algorithms, introducing artifacts that are valuable in forensic analysis.

In the latter case, each pixel can be associated to a single color. As a matter of fact, it is necessary to estimate the missing colors by interpolation methods. Such activity introduces several artifacts that are peculiar to the specific interpolation method. Remembering that each camera vendor implements its own demosaicing strategy (often strategy changes from model to model, from firmware version to version), it is possible to analyze such artifacts to identify camera model (not instance like for PRNU).

Let us consider the interpolated signal

$$\hat{I}_d(x,y) = \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v}\, I(x+u, y+v)$$

and the residual

$$r(x,y) = I(x,y) - \sum_{u=-m}^{m} \sum_{v=-m}^{m} \alpha_{u,v}\, I(x+u, y+v) \quad (*)$$

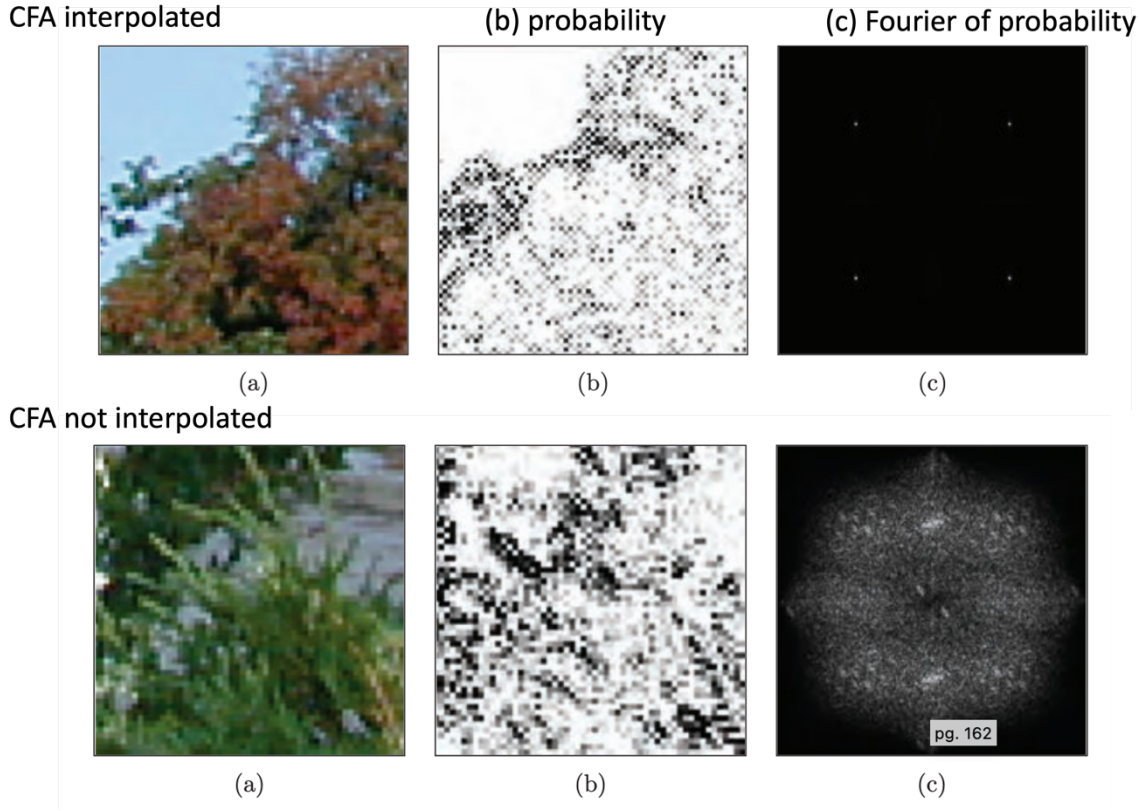
The likelihood of observing a pixel generated by weights $\alpha = [\alpha_{u,v}]$ is

$$w(x,y) = \frac{e^{-r^2/\sigma}}{e^{-r^2/\sigma} + \delta}$$

where δ is uniform probability of observing a different model or authentic pixels. It is possible to compute this with an EM estimation where the E-step is the residual estimation in (*), i.e., find the α such that residual is minimized, and the M-step is performed by the equation

$$E(\boldsymbol{\alpha}) = \sum_{x=m+1}^{n-m} \sum_{y=m+1}^{n-m} w(x,y)\, r(x,y)$$

where weights w are computed to minimize $E(\boldsymbol{\alpha})$ (give more weights whenever you are sure you have found the correct interpolation coefficients).



Compression traces

Compression can be categorized into:
Lossless Compression: ensures perfect reconstruction without distortion.
Lossy Compression: trades fidelity for reduced data size, focusing on perceptual relevance (e.g., JPEG, MPEG formats).

Lossy compression leaves artifacts on the reconstructed image/video/audio file that are dependent on the specific implementation. Even if the coding architectures follows the specifications of a coding standard, some implementation aspects are non-normative, i.e., they are not defined by the standard protocol and can be implemented as the developer desires. Since some of these elements are deeply related to the efficiency of the compression (i.e., the ability of coding the originally-acquired samples into a limited storage file with minimum distortion), most vendors implement (and patent) their own compression strategy (quantization policy, motion estimation algorithm, denoising, etc.) which becomes a distinctive element of that specific implementation. In this paragraph we overview some of the different coding traces that can be used for device identification.

Coding tools for image/video/audio compression

Most of the existing coding architectures build on top of coding tools originally designed for images. The JPEG standard is, by far, the most widely adopted coding technique for still images and many of its principles are reused for the compression of video signals. A JPEG codec converts color images into a suitable color, and processes each color component independently. The encoder operates according to three main steps:

- The image is divided into non-overlapping 8 × 8 pixel blocks, which are then transformed by Discrete Cosine Transform (DCT).

- DCT coefficients are uniformly quantized into integer levels by means of different quantization steps that can be varying according to the spatial frequencies (horizontal and vertical coordinates of the coefficient)
- levels are lossless coded into a binary bitstream by means of entropy coding (in the case of JPEG, Huffman coding tables)

Other compression standards, like JPEG2000, avoid block partitioning by adopting a more general image-wise transform (Wavelet transform).

Video coding architectures are more complex than still image coding, incorporating block-wise transform coding (as in JPEG) alongside additional tools such as spatial and temporal prediction, in-loop filtering, and image interpolation. These architectures often use varying transforms within a single standard. A typical video encoder divides a video sequence into frames, which are further split into pixel blocks. Each block is subtracted from a prediction (leveraging spatial/temporal correlation): indeed, predictive techniques are used reduce redundancy by approximating the blocks in the current frame with some equivalent blocks in the previously-decoded information. Motion-compensated prediction (MC-P) calculates differences between adjacent frames, encoding only the residual signal, which is then encoded through steps akin to JPEG, with quantization steps and transform characteristics varying by standard.

All these operations introduce some artifacts that can be used for codec identification.

Quantization artifacts

Quantization is a non-invertible operation and it is the main source for information loss. Thus, it leaves characteristic footprints, which depend on the chosen quantization steps and quantization strategy. Therefore, the analysis of
coding-based footprints might be leveraged to:

- infer details about the encoder (e.g. coding standard, coding parameters, non-normative tools);
- assess the quality of a sequence in a no-reference framework; or
- study the characteristics of the channel used to transmit the sequence.

In addition, block-wise processing introduces an artificial partition of the coded frame, which is further enhanced by the following processing steps. Unlike JPEG, the actual partitioning strategy is not fixed, as it depends on the specifications of coding standard and on the adopted rate–distortion optimization policy. Therefore, blockiness artifacts can be used to infer information about the adopted
codec.

Detecting quantization parameters or recompression traces is largely useful for several operations, like identifying the originating device or repository on an image/video/audio. Indeed, images uploaded to social media are often recompressed, altering statistical patterns, which can be exploited for forensic analysis. Moreover, the recompression of parts of an image with respect to others can often be leveraged to detect the presence or tampering and alterations.
Such operations can be performed analyzing the statistics of DCT coefficients with ML techniques or verifying some statistical properties, e.g., Benford's law.

Prediction artifacts

Redundancy in video frames is minimized using predictive techniques. Motion-compensated prediction (MC-P) calculates differences between adjacent frames, encoding only the residual signal.
Different codec implementations may adopt diverse spatial or temporal prediction strategies, according to rate–distortion requirements and computational constraints. The identification of

the adopted motion vectors and coding modes provides relevant footprints that can be exploited by the forensic analyst, e.g. to validate the originating devices.

Notes about scanner and printer forensics

Among the possible physics-related traces that we can use in multimedia forensics, it is worth including a separate mention for those referred to printers and scanners. Indeed, such devices that are processed printed materials bring forth many revealing elements that account for the identity of the specific device[3]. In this activity, we need to distinguish between

- **intrinsic signatures**, which are device-specific features from manufacturing defects or mechanical operations,
- **extrinsic signatures**, embedded actively during printing to include metadata like printer serial numbers.

Such traces can be extracted and used to identify laser jet and ink jet printers as well.

Traces for Identifying Laser-Jet Printers

Laser-jet printers use electrophotographic (EP) technology, which produces identifiable artifacts stemming from their mechanical and optical components.

1. Banding Patterns

Such traces are due to the fluctuations in the angular velocity of the optical photoconductor (OPC) drum or gear transmission errors.
They appear as alternating light and dark bands perpendicular to the process direction.
Frequency-domain analysis of large midtone regions in the printed document helps isolate banding frequencies. These frequencies are correlated with specific gear teeth errors or OPC drum velocities.
Banding frequencies are unique to printer models, making them effective for device characterization.

2. Texture Features

Such footprints are due to electromechanical imperfections during toner application. They appear as variations in toner distribution create unique textural features that can be analyzed using a *gray-level co-occurrence matrix (GLCM)* and classified with machine learning methods.
Such tracesa re useful to discriminates between printers, even under text-only conditions.

3. Toner and Dot Gain Characteristics

Sharpness, toner distribution, and dot size variations under different conditions are present because of the physical and chemical properties of the toner and fusing mechanisms.
Optical profiling can reveal how toner protrudes above the paper surface.

4. Laser Pulse Modulation

Dot placement and size are usually controlled using laser intensity modulation. In such operation, it is possible to encode additional signals (e.g., sinusoidal patterns) for forensic tracing. Frequency analysis can recover these embedded signals for identification.
In this way, we are combining intrinsic and extrinsic methods for a more robust device tracking.

Traces for Identifying Ink-Jet Printers

[3] Chiang et al. “Printer and Scanner Forensics”, IEEE Sig.Proc. Mag., 2009

Ink-jet printers rely on liquid ink dispensed through nozzles, producing distinct artifacts based on their operation and print modes. As a matter of fact, several traces can be retrieved

1. Dot Asymmetry and Placement
Dot shape becomes asymmetrical at higher print speeds, with "tails" or "satellites" trailing the main dot. Moreover, misalignments in nozzle arrays cause characteristic placement errors.

2. Swath and Pass Modes
Horizontal bands (swaths) can be printed by the printhead during each pass. In such, process, overlapping or misaligned swaths indicate specific print modes (e.g., single-pass or multi-pass).

3. Nozzle and Fluid Dynamics
Ink flow through the printhead and nozzle geometry. Structural deviations in the nozzle plate create "toe-in" or dimpled effects on dot placement.

4. Paper Handling Artifacts
Such traces are depending on the mechanical processes like paper advancement and gear operation. Spur marks left by paper conveyance gears can be detected using infrared or oblique lighting. Patterns of spur marks are unique to printer models.

Comparative Analysis

Feature	Laser-Jet Printers	Ink-Jet Printers
Banding	High-frequency, gear-related	Not applicable
Dot Asymmetry	Minimal due to toner-based process	Significant, influenced by speed
Texture Features	Toner uniformity and sharpness	Dependent on nozzle geometry
Paper Handling Artifacts	Minor	Prominent, due to spur marks
Swath/Pass Artifacts	Not applicable	Directional variations evident